

Extrema-Frame Segmentation of Meibography Images: Formal Analysis and an Output-Preserving Linear-Time Reformulation

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Abstract

We study a deterministic row-wise segmentation procedure originally developed for meibography images and expose the mathematical structure underlying its extrema selection, frame construction, and pixel-classification mechanisms. After a region of interest is specified, each active image row is represented as a one-dimensional intensity signal. A contrast parameter Δ controls bilateral directional tests whose outcomes admit an exact prominence-based characterization. We show that the apparently stateful peak-selection mechanism decomposes into a static sequence of prominence-qualified candidates followed by a deterministic refinement whose independent effect is confined to equal-height recurrence. In the absence of intensity ties, canonical peak selection reduces exactly to thresholding the bilateral prominence score.

The accepted peaks and running minima generate a unique four-type frame representation of each row. We prove complete row coverage, shared canonical boundaries, uniqueness, and a merge-only hierarchy with respect to Δ : increasing the contrast parameter can remove peaks and frame boundaries but cannot relocate surviving boundaries. Consequently, the family of canonical interval decompositions forms a laminar coarsening hierarchy with only linearly many distinct structural levels.

We further show that the original regular-frame pixel rule is exactly equivalent to a nearest-anchor decision, a local midpoint classifier, and, after local contrast normalization, the universal rule $\alpha(x) > 1/2$. Boundary frames admit the unified normalized criterion $\alpha(x) > \max\{1/2, \Delta/D\}$, yielding an explicit three-regime contrast characterization. These results lead to specification correctness, determinism, row separability, local auditability, and photometric-equivariance properties for the complete segmentation operator.

Finally, we show that the per-position bidirectional searches of the canonical procedure can perform quadratic total work in the worst case on a row. By replacing those searches with two augmented monotone-stack passes that recover the same nearest-strictly-greater boundaries and bounded minima, while retaining the original state and frame semantics, we derive an exact output-preserving linear-time reformulation. The accelerated procedure produces the same accepted peaks, the same frames, and the same segmentation mask in $\Theta(n)$ time per processed row and $\Theta(|\Omega|)$ time for an explicit ROI mask under the stated model.

Keywords: deterministic image segmentation; meibography; peak prominence; extrema-frame decomposition; local adaptive thresholding; monotone stack; interpretable segmentation; algorithm analysis.

1 Introduction

The procedure studied in this work originates from a web-based software system developed for meibography processing in 2019. The original workflow proceeds from region-of-interest

specification to row-wise intensity processing, extrema detection, frame construction, and pixel classification. The present paper isolates the pre-revision extrema-frame segmentation core of that work and treats it as a precisely defined algorithmic object. Later width modification, gap filling, brightness-related revision mechanisms, and post-segmentation manual editing are deliberately excluded from the formal operator analyzed here.

Although the reference procedure is operationally straightforward, its mathematical behavior is not immediately apparent from the procedural description. The directional extrema search may extend over a variable distance, accepted peaks interact with a running-minimum state, adjacent frames share extrema, and different frame types employ superficially different pixel-selection rules. Several natural questions therefore arise. What signal property is actually measured by the bilateral extrema test? What independent role is played by the running state? Does the frame representation have a unique structural form? How does that representation evolve as Δ changes? Can the apparently different frame rules be expressed through a common classifier? Finally, is the per-position bidirectional search necessary for preserving the segmentation semantics, or can the same directional information be recovered more efficiently?

The analysis reveals a simpler structure beneath the original stateful implementation. The bilateral directional tests can be summarized by a scalar prominence score for each candidate position. Once the prominence-qualified candidate sequence is known, the running-minimum state can be eliminated from peak-set determination except for a deterministic refinement of equal-height candidate recurrence. In the tie-free case, canonical peak selection therefore reduces exactly to prominence thresholding. The resulting accepted peaks uniquely determine a shared-boundary frame representation, and increasing Δ induces a merge-only hierarchy in which existing boundaries may disappear but surviving boundaries do not drift.

A second simplification occurs at the pixel-classification stage. The original regular-frame rule compares the intensity distance of a pixel from a local minimum with its distance from the associated peak. We show that this comparison is exactly equivalent to thresholding the local intensity midpoint and, after normalization by the local minimum-to-peak contrast, to the universal decision $\alpha > 1/2$. Boundary frames admit the closely related rule $\alpha > \max\{1/2, \Delta/D\}$. Thus the canonical segmentation can be interpreted as an extrema-defined local normalization scheme followed by explicit deterministic classification rather than as a single global grayscale threshold.

The canonical operator contains no learned weights, random initialization, or stochastic inference mechanism. Once the image, ROI, and contrast parameter are fixed, every accepted extremum, frame boundary, local threshold, and pixel label is uniquely determined by explicit comparisons. Individual output decisions can therefore be reconstructed from the input values and the corresponding frame anchors. These properties concern reproducibility and specification-level correctness; they do not by themselves constitute a claim of external clinical accuracy.

The same structural analysis exposes the source of the computational inefficiency of the canonical directional procedure. The outer procedure traverses the active positions of a row once, but each current position initiates a leftward and a rightward search, and search intervals initiated from different positions may overlap substantially. We show that the total directional work can be $\Theta(n^2)$ in the worst case. The nearest strictly greater blockers and the bounded minima that determine all directional outcomes can instead be recovered by two augmented monotone-stack passes. Retaining the canonical running state and frame interpretation then yields an exact output-preserving linear-time formulation.

The contributions of this work are sixfold. First, we give a mathematical specification of the canonical extrema-frame segmentation procedure and separate its static directional geometry from its state-dependent behavior. Second, we derive a prominence-based characterization of the bilateral extrema test and show that, once prominence-qualified candidates are known, the independent effect of the running state is confined to equal-height recurrence; in particular, tie-free peak selection becomes state-free prominence thresholding. Third, we prove that accepted

peaks uniquely determine a four-type shared-boundary frame representation and that increasing Δ produces a merge-only laminar hierarchy with no boundary drift. Fourth, we show that the original frame-distance rule is exactly equivalent to a locally normalized classifier with universal regular-frame threshold $\alpha > 1/2$, and we derive a unified normalized rule and three contrast regimes for boundary frames. Fifth, we establish specification correctness, deterministic output, row separability, decision auditability, and several invariance and parameter-structure properties, while also identifying limits such as non-monotonicity of the final mask across structural transitions. Sixth, we derive an exact output-preserving augmented-stack reformulation that reduces the worst-case running time from quadratic to linear per row without changing the accepted peaks, frame representation, or final segmentation mask.

The scope of the present work is algorithmic. No claim of clinical ground-truth validation is made here. The historical software lineage is documented in section 9, but the output-preservation theorem refers specifically to the canonical pre-revision operator defined in section 2, not to later revision layers present in the archived software.

2 Computational Setting and Canonical Algorithm

2.1 Image, ROI, and row signals

Let

$$I : \{0, \dots, W - 1\} \times \{0, \dots, H - 1\} \rightarrow \mathcal{I}$$

be a grayscale image. In the 8-bit setting, $\mathcal{I} = \{0, \dots, 255\}$. Let

$$\Omega \subseteq \{0, \dots, W - 1\} \times \{0, \dots, H - 1\}$$

be the region of interest (ROI). For row y , define the active domain

$$D_y = \{x : (x, y) \in \Omega\}.$$

Assumption 1 (Admissible row slices). Every nonempty horizontal ROI slice is a contiguous integer interval:

$$D_y = [S_y, E_y] \cap \mathbb{Z}.$$

This assumption isolates the row operator studied below. An extension to disconnected horizontal slices would require an explicit component-reset convention and is outside the present analysis.

For a nonempty active row, define

$$P_y(x) = I(x, y), \quad x \in D_y.$$

When one row is considered, the subscript is suppressed and we write

$$P : [S, E] \cap \mathbb{Z} \rightarrow \mathbb{R}, \quad n = E - S + 1.$$

2.2 Contrast parameter and canonical comparison conventions

Let $\Delta > 0$ be the contrast parameter. The historical implementation used a default value corresponding to VAR=10; the theory treats Δ as a variable parameter.

The canonical operator uses the following strict conventions throughout:

- (i) a directional blocker satisfies $P(j) > P(p)$;
- (ii) a successful directional drop satisfies $P(p) - P(j) > \Delta$;
- (iii) the running minimum is updated only under a strict decrease;
- (iv) regular and boundary pixel-classification inequalities are strict.

Consequently, an equality between two candidate minima preserves the earlier coordinate, giving a leftmost-minimizer tie rule.

2.3 Directional tests

For a current position p , the rightward test examines $p + 1, p + 2, \dots$ in order. It returns false when a sample strictly greater than $P(p)$ is reached before a successful contrast drop; it returns true when a sample j satisfying $P(p) - P(j) > \Delta$ is reached first; and it returns false if the row boundary is reached without success. Denote the result by $R_\Delta(p)$. Define $L_\Delta(p)$ symmetrically.

Equivalently, define nearest strictly greater blockers

$$g_R(p) = \min\{j > p : P(j) > P(p)\},$$

with sentinel $E + 1$ when no such index exists, and

$$g_L(p) = \max\{j < p : P(j) > P(p)\},$$

with sentinel $S - 1$ when none exists. Define directional depths

$$d_R(p) = \max_{p < j < g_R(p)} (P(p) - P(j)), \quad d_L(p) = \max_{g_L(p) < j < p} (P(p) - P(j)),$$

where the maximum over an empty set is defined as $-\infty$. Then

$$R_\Delta(p) \iff d_R(p) > \Delta, \quad L_\Delta(p) \iff d_L(p) > \Delta.$$

2.4 Running-minimum state and peak acceptance

Before any peak has been accepted, the running-minimum position at p is

$$B(p) = \text{leftmostargmin}_{S \leq j \leq p} P(j).$$

If q is the latest accepted peak, then

$$B(p) = \text{leftmostargmin}_{q < j \leq p} P(j).$$

The canonical peak predicate is

$$\text{Peak}_\Delta(p) \iff L_\Delta(p) \wedge R_\Delta(p) \wedge (P(p) - P(B(p)) > \Delta). \quad (1)$$

An accepted peak resets the running-minimum state.

2.5 Frame construction

Let accepted peaks be

$$p_1 < \dots < p_k.$$

For $k \geq 1$, define canonical minima

$$b_1 = \text{leftmostargmin}_{S \leq x \leq p_1} P(x),$$

$$b_i = \text{leftmostargmin}_{p_{i-1} < x \leq p_i} P(x), \quad 2 \leq i \leq k,$$

and

$$b_{k+1} = \text{leftmostargmin}_{p_k < x \leq E} P(x).$$

The regular frames are

$$F_i = (b_i, p_i, b_{i+1}), \quad 1 \leq i \leq k.$$

If $b_1 > S$, the residual prefix $[S, b_1]$ is a left boundary frame. If $b_{k+1} < E$, $[b_{k+1}, E]$ is a right boundary frame. If $k = 0$, the entire row $[S, E]$ is a whole-row boundary frame.

2.6 Canonical frame interpretation

For a regular frame $F = (L, p, R)$, a left-half pixel $x \in [L, p]$ is selected iff

$$P(x) - P(L) > P(p) - P(x), \quad (2)$$

and a right-half pixel $x \in [p, R]$ is selected iff

$$P(x) - P(R) > P(p) - P(x). \quad (3)$$

For a boundary frame $B = [a, b]$, let

$$m_B = \min_{a \leq j \leq b} P(j), \quad M_B = \max_{a \leq j \leq b} P(j).$$

A boundary pixel $x \in B$ is selected iff both

$$P(x) - m_B > M_B - P(x) \quad (4)$$

and

$$P(x) - m_B > \Delta \quad (5)$$

hold.

Let $G_\Delta(F)$ denote the selected positions in frame F . The canonical row segmentation is

$$\mathcal{S}_\Delta(P) = \bigcup_{F \in \mathcal{F}_\Delta(P)} G_\Delta(F),$$

and the global canonical segmentation operator is

$$\mathcal{A}_\Delta^{\text{can}}(I, \Omega) = \bigcup_{y: D_y \neq \emptyset} \{(x, y) : x \in \mathcal{S}_\Delta(P_y)\}. \quad (6)$$

2.7 Canonical pseudocode

Listing 1: Canonical extrema-frame segmentation of one row.

```

Input: P[S..E], contrast Delta > 0
Output: selected positions G

frames <- empty
runningMin <- undefined

for p = S,...,E:
  if runningMin is undefined or P[p] < P[runningMin]:
    runningMin <- p

  leftOK <- LeftDirectionalTest(P,p,Delta)
  rightOK <- RightDirectionalTest(P,p,Delta)

  if leftOK and rightOK and P[p]-P[runningMin] > Delta:
    if a previous regular frame exists:
      append runningMin as its right minimum
    start a new regular frame [runningMin,p]
    runningMin <- undefined

complete the last regular frame and residual boundary frame(s),
or create [S,E] if no regular peak was accepted
interpret all frames by the canonical rules
return the union G

```

The canonical operator ends with the frame-based classification above. Width modification, thickening or thinning, gap filling, connected-region correction, brightness-related revision procedures, and manual post-segmentation editing are not components of $\mathcal{A}_\Delta^{\text{can}}$.

3 Prominence-Controlled Extrema Detection

3.1 Bilateral prominence score

Define

$$\pi(p) = \min\{d_L(p), d_R(p)\}.$$

The static bilateral candidate set is

$$\mathcal{C}_\Delta(P) = \{p : \pi(p) > \Delta\}.$$

Lemma 1 (Bilateral-prominence equivalence). *For every position p ,*

$$L_\Delta(p) \wedge R_\Delta(p) \iff \pi(p) > \Delta.$$

Proof. By construction, each directional predicate is true exactly when the largest visible drop before the corresponding strictly greater blocker exceeds Δ . Taking the conjunction is therefore equivalent to requiring the smaller of the two directional depths to exceed Δ . $\square$

For equal-height positions $a < b$, define the inter-position valley depth

$$V(a, b) = P(a) - \min_{a \leq j \leq b} P(j), \quad P(a) = P(b).$$

If $a < b < c$ have the same height, then

$$V(a, c) = \max\{V(a, b), V(b, c)\}. \tag{7}$$

Theorem 1 (Prominence-state decomposition and adjacent-candidate characterization). *Let*

$$\mathcal{C}_\Delta(P) = \{c_1 < \dots < c_m\}.$$

If $m > 0$, c_1 is accepted. For $i \geq 2$,

$$c_i \in \mathcal{P}_\Delta(P) \iff \begin{cases} P(c_i) \neq P(c_{i-1}), \\ P(c_i) = P(c_{i-1}) \text{ and } V(c_{i-1}, c_i) > \Delta. \end{cases} \quad \text{or}$$

Thus, once the bilateral candidate sequence is known, peak-set determination requires only adjacent candidate heights and, for equal-height neighbors, their intervening valley depth.

Proof. The first bilateral candidate has a successful left witness whose intensity is more than Δ below the candidate. The running minimum over the candidate prefix is no higher than that witness, so the state gate in (1) is also satisfied.

Consider c_i , $i \geq 2$. A bilateral candidate can be rejected by the state gate only when its height equals that of the latest accepted peak. If $P(c_i) \neq P(c_{i-1})$, then either c_{i-1} was accepted, in which case the latest accepted height differs from $P(c_i)$, or c_{i-1} was rejected, in which case its height equals the latest accepted height and again differs from $P(c_i)$. Hence c_i is accepted.

Now suppose $P(c_i) = P(c_{i-1}) = H$. If c_{i-1} was accepted, the state gate for c_i is exactly $V(c_{i-1}, c_i) > \Delta$. If c_{i-1} was rejected, let q be the latest accepted peak before it. Rejection implies $P(q) = H$ and $V(q, c_{i-1}) \leq \Delta$. By (7),

$$V(q, c_i) = \max\{V(q, c_{i-1}), V(c_{i-1}, c_i)\},$$

so $V(q, c_i) > \Delta$ holds exactly when $V(c_{i-1}, c_i) > \Delta$. This is the canonical state condition for c_i . $\square$

Corollary 1 (Tie-free simplification). *If all row intensities are distinct, then*

$$\mathcal{P}_\Delta(P) = \mathcal{C}_\Delta(P) = \{p : \pi(p) > \Delta\}.$$

Corollary 2 (Plateau canonicalization). *A constant-intensity plateau contributes at most one accepted peak. If a point from the plateau is accepted, it is the first prominence-qualified candidate on that plateau.*

Define the canonical survival scale of a position by

$$\lambda(p) = \sup(\{\Delta > 0 : p \in \mathcal{P}_\Delta(P)\} \cup \{0\}).$$

Always $0 \leq \lambda(p) \leq \pi(p)$, and in the tie-free case $\lambda(p) = \pi(p)$.

Theorem 2 (Accepted-peak monotonicity). *If $0 < \Delta_1 < \Delta_2$, then*

$$\mathcal{P}_{\Delta_2}(P) \subseteq \mathcal{P}_{\Delta_1}(P).$$

Equivalently,

$$\mathcal{P}_\Delta(P) = \{p : \lambda(p) > \Delta\}.$$

Proof. Let p be accepted at Δ_2 . Since $\pi(p) > \Delta_2 > \Delta_1$, p is also a bilateral candidate at Δ_1 . Suppose, for contradiction, that p is rejected at Δ_1 . By theorem 1, the latest lower-threshold accepted peak r before p must satisfy $P(r) = P(p)$ and $V(r, p) \leq \Delta_1$.

Because p remains a bilateral candidate at the larger threshold, a same-height point r separated from p by a valley of depth at most $\Delta_1 < \Delta_2$ cannot be bypassed by a strictly greater blocker between them; the directional witnesses certifying p at Δ_2 also certify r as a Δ_2 bilateral candidate. Let q be the latest Δ_2 -accepted peak before p . Since r is not accepted at Δ_2 , the adjacent-candidate characterization implies $P(q) = P(r) = P(p)$ and $V(q, r) \leq \Delta_2$. Yet acceptance of p at Δ_2 requires $V(q, p) > \Delta_2$. By the equal-height composition identity,

$$V(q, p) = \max\{V(q, r), V(r, p)\} \leq \Delta_2,$$

a contradiction. □

Proposition 1 (Peak separation). *Consecutive accepted peaks satisfy $p_{i+1} - p_i \geq 2$. Hence*

$$|\mathcal{P}_\Delta(P)| \leq \left\lfloor \frac{n-1}{2} \right\rfloor.$$

Proof. An accepted peak needs a successful directional witness on each side. In particular, two adjacent positions cannot both be accepted because each would act as either a non-lower immediate neighbor or a strictly greater blocker for the other's required directional condition. The counting bound follows by packing positions with a gap of at least one index between consecutive accepted peaks. □

Proposition 2 (No fixed-radius local characterization). *For every fixed radius r , there exist two row signals that agree on the radius- r neighborhood of a position p but differ in whether p satisfies the bilateral candidate condition at a prescribed Δ .*

Proof. Place identical values in the radius- r neighborhood of p and make the visible descent on one side remain shallower than Δ throughout that neighborhood. Outside the neighborhood, extend one signal with a sufficiently low witness before any strictly greater blocker, and extend the other with a strictly greater blocker first. The local neighborhoods coincide while the directional predicate differs. □

4 Extrema-Frame Decomposition

Theorem 3 (Unique canonical four-type frame decomposition). *For every finite admissible row signal P and fixed $\Delta > 0$, the canonical construction generates a unique frame family $\mathcal{F}_\Delta(P)$. If at least one peak is accepted, the regular frames are $F_i = (b_i, p_i, b_{i+1})$, possibly supplemented by a left boundary frame $[S, b_1]$ and a right boundary frame $[b_{k+1}, E]$. If no peak is accepted, the family consists of the whole-row boundary frame $[S, E]$. The frame domains cover the row, and distinct frame interiors are disjoint.*

Proof. The accepted peak sequence is unique by the deterministic predicate (1). The strict running-minimum update selects the leftmost minimizer on each interval between consecutive accepted peaks, so each b_i is unique. The same minimum that closes one regular frame is the left minimum of the next, hence adjacent regular frames share exactly one canonical minimum anchor. The four resulting frame types are summarized schematically in fig. 1. The residual prefix and suffix, when nonempty, are the two boundary frame types; with no accepted peak, the whole row is the fourth type. Their consecutive domains cover $[S, E]$ and may overlap only at shared endpoints. $\square$

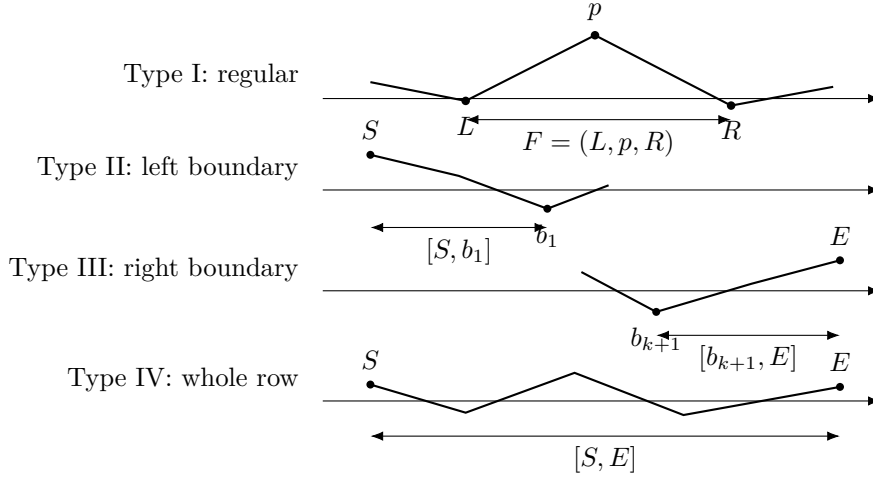


Figure 1: Schematic of the four canonical frame types. A regular frame is anchored by a left minimum, an accepted peak, and a right minimum. Residual prefix and suffix intervals form the two boundary-frame types; if no peak is accepted, the whole row is a single boundary frame. The drawing is structural and not to intensity scale.

Proposition 3 (Peak set determines frame structure). *For fixed P ,*

$$\mathcal{P}_{\Delta_1}(P) = \mathcal{P}_{\Delta_2}(P) \iff \mathcal{F}_{\Delta_1}(P) = \mathcal{F}_{\Delta_2}(P).$$

Proof. If the peak sequence is the same, the intervals over which all canonical minima are taken are the same; the leftmost-minimum tie rule then gives identical minima and frames. Conversely, each regular frame contains its unique accepted peak coordinate, and the no-peak case is represented by the whole-row boundary frame. $\square$

Let $B_\Delta(P)$ be the set containing S, E and all canonical minimum-boundary coordinates induced by $\mathcal{F}_\Delta(P)$.

Theorem 4 (Peak-determined merge-only frame hierarchy). *If $0 < \Delta_1 < \Delta_2$, then*

$$B_{\Delta_2}(P) \subseteq B_{\Delta_1}(P).$$

Thus a surviving canonical boundary never changes coordinate as Δ increases. Every higher- Δ cell is a union of consecutive lower- Δ cells; fig. 2 illustrates this merge-only evolution.

Proof. By theorem 2, every peak surviving at Δ_2 also occurs at Δ_1 . Consider consecutive surviving peaks $q < r$ at the larger threshold. At the smaller threshold there may be extra peaks between them, partitioning $(q, r]$ into consecutive subintervals. The larger-threshold boundary between q and r is the leftmost minimum over the union of these subintervals. The minimum value over a union of consecutive intervals is attained at one of their constituent minima; the leftmost tie rule selects the earliest such constituent minimum. Hence the larger-threshold boundary already appears at the smaller threshold. Prefix and suffix boundaries obey the same argument. $\square$

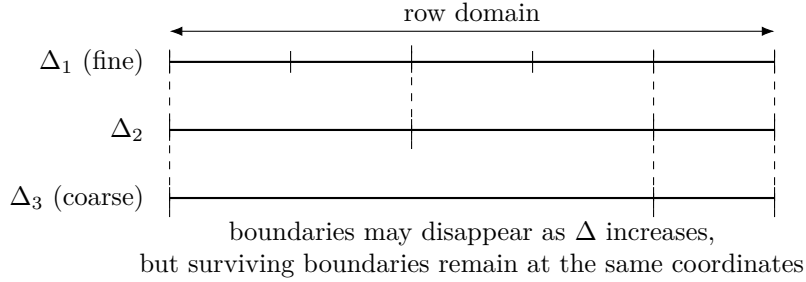


Figure 2: Merge-only Δ -indexed frame hierarchy. Higher contrast thresholds remove structural boundaries and merge adjacent cells; a surviving boundary does not drift horizontally.

Using a fixed half-open convention for the cells induced by $B_\Delta(P)$ gives a partition $\Pi_\Delta(P)$.

Corollary 3 (Laminar interval hierarchy). *The family $\bigcup_{\Delta>0} \Pi_\Delta(P)$ is laminar: for any two cells A, B from arbitrary parameter levels,*

$$A \cap B = \emptyset, \quad A \subseteq B, \quad \text{or} \quad B \subseteq A.$$

Corollary 4 (Number of distinct frame structures). *If $K_{\max} = \max_{\Delta>0} |\mathcal{P}_\Delta(P)|$, then*

$$\#\{\text{distinct canonical frame structures}\} \leq K_{\max} + 1 \leq \left\lfloor \frac{n-1}{2} \right\rfloor + 1.$$

5 Frame Interpretation as Local Normalized Classification

Lemma 2 (Regular-frame contrast and peak boundedness). *Let $F = (L, p, R)$ be a canonical regular frame. Then*

$$P(p) - P(L) > \Delta, \quad P(p) - P(R) > \Delta,$$

and

$$\begin{aligned} P(L) &\leq P(x) \leq P(p) & (L \leq x \leq p), \\ P(R) &\leq P(x) \leq P(p) & (p \leq x \leq R). \end{aligned}$$

Proof. The left contrast inequality is the state-gate condition at the moment p is accepted, because L is the current running minimum. For the right contrast, let $r > p$ be a right directional witness with $P(p) - P(r) > \Delta$. Before the first such witness the reset running-minimum gate cannot exceed Δ , so no new accepted peak can close the frame before r is encountered. Hence r lies within the interval from which R is selected, and $P(R) \leq P(r)$.

For the upper bound on the left half, suppose some $x \in [L, p)$ has $P(x) > P(p)$, and choose a maximum-height such x . The left minimum L gives a drop larger than Δ , while a right witness

for p is also visible from x because no intervening sample exceeds $P(x)$. The running-minimum gate for x is no harder than the drop to L . Thus x would be accepted before p , contradicting the frame construction. The right-half upper bound is symmetric after the reset: a successful right witness of p occurs before the first sample higher than $P(p)$, and any higher maximum before R would then satisfy both directional tests and the state gate, forcing an intervening accepted peak. The lower bounds follow from the definitions of L and R as the corresponding minima. $\square$

Lemma 3 (Shared-anchor decision consistency). *A canonical minimum shared by two adjacent regular frames is rejected by both frame interpretations. A minimum shared between a regular frame and a boundary frame is likewise rejected by both.*

Proof. At a regular minimum anchor, the distance to that minimum is zero while the distance to the corresponding peak is strictly positive by lemma 2, so the strict regular selection inequality fails. In a boundary frame the same minimum has contrast zero above the frame minimum and therefore fails (5). $\square$

Theorem 5 (Exact local normalized classification). *Let U be either half of a canonical regular frame, with minimum intensity m_U , peak intensity M_U , and contrast $D_U = M_U - m_U > 0$. Define*

$$\alpha_U(x) = \frac{P(x) - m_U}{D_U}.$$

For every $x \in U$, the following are equivalent:

$$P(x) - m_U > M_U - P(x),$$

$$|P(x) - M_U| < |P(x) - m_U|,$$

$$P(x) > \frac{m_U + M_U}{2},$$

and

$$\alpha_U(x) > \frac{1}{2}.$$

Proof. By lemma 2, $m_U \leq P(x) \leq M_U$, so the signed differences equal the corresponding absolute distances. Rearranging the strict distance inequality gives $2P(x) > m_U + M_U$. Subtracting m_U and dividing by $D_U > 0$ yields $\alpha_U(x) > 1/2$. $\square$

Corollary 5 (Fixed-frame regular classification). *Once a regular frame is fixed, its selected pixel set is independent of Δ .*

Theorem 6 (Unified boundary classification). *Let B be a canonical boundary frame, with*

$$m_B = \min_{x \in B} P(x), \quad M_B = \max_{x \in B} P(x), \quad D_B = M_B - m_B.$$

A boundary pixel is selected exactly when

$$P(x) > m_B + \max \left\{ \frac{D_B}{2}, \Delta \right\}.$$

If $D_B > 0$ and $\alpha_B(x) = (P(x) - m_B)/D_B$, this is equivalent to

$$\alpha_B(x) > \max \left\{ \frac{1}{2}, \frac{\Delta}{D_B} \right\}.$$

If $D_B = 0$, the selected boundary set is empty.

Proof. The midpoint condition (4) is equivalent to $P(x) > m_B + D_B/2$, while (5) is equivalent to $P(x) > m_B + \Delta$. Both hold exactly above the maximum of the two thresholds. Normalization by $D_B > 0$ gives the second form. For $D_B = 0$, no pixel can have contrast greater than $\Delta > 0$ above the minimum. $\square$

Corollary 6 (Boundary contrast regimes). *For $D_B > 0$,*

$$G_\Delta(B) = \begin{cases} \emptyset, & D_B \leq \Delta, \\ \{x : \alpha_B(x) > \Delta/D_B\}, & \Delta < D_B < 2\Delta, \\ \{x : \alpha_B(x) > 1/2\}, & D_B \geq 2\Delta. \end{cases}$$

Moreover, $G_\Delta(B) \neq \emptyset$ iff $D_B > \Delta$.

The regular and boundary normalized thresholds are compared visually in fig. 3.

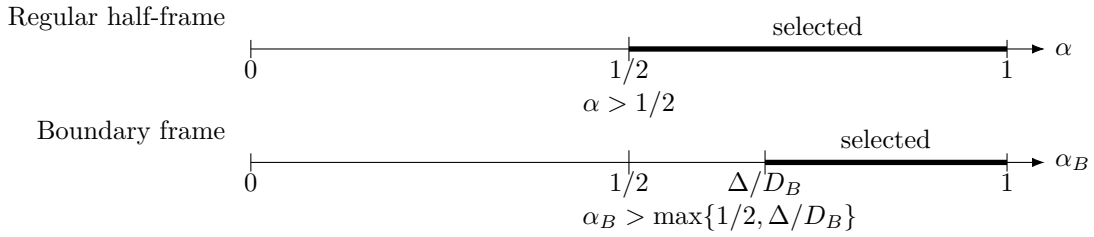


Figure 3: Normalized interpretation of the canonical pixel rules. A regular half-frame always uses the universal midpoint threshold $\alpha > 1/2$. A boundary frame uses the larger of the midpoint threshold and the explicit contrast safeguard Δ/D_B ; the illustrated ordering $\Delta/D_B > 1/2$ corresponds to the intermediate-contrast regime.

6 Global Segmentation Operator and Correctness

Define the explicit frame specification

$$\text{Spec}_\Delta(P) = \bigcup_{F \in \mathcal{F}_\Delta(P)} \{x \in \text{dom}(F) : C_\Delta(x; F)\},$$

where C_Δ is the normalized regular predicate of theorem 5 or the boundary predicate of theorem 6, according to frame type.

Theorem 7 (Specification-correct deterministic segmentation). *For every finite admissible row P ,*

$$\mathcal{S}_\Delta(P) = \text{Spec}_\Delta(P).$$

For every finite image I , admissible ROI Ω , and $\Delta > 0$, there exists a unique mask

$$G = \mathcal{A}_\Delta^{\text{can}}(I, \Omega).$$

Proof. Every pixel added by the canonical regular-frame procedure satisfies (2) or (3), which is equivalent to the normalized regular predicate by theorem 5. Every pixel added by a boundary frame satisfies exactly (4)–(5), equivalent to the boundary specification by theorem 6. This gives soundness. Conversely, every pixel satisfying the explicit specification belongs to a canonical frame, every canonical frame is processed, and the same equivalences convert the specification predicate back to the implemented canonical inequalities; this gives completeness.

Uniqueness follows because the input signal and Δ uniquely determine all strict directional comparisons, the running minimum under the fixed leftmost tie convention, the accepted peak

sequence, the frame family, and every Boolean frame decision. Shared endpoints cannot create contradictory decisions by the shared-anchor lemma. Row-wise union then gives a unique image mask. $\square$

Corollary 7 (Complete ROI classification). *Every ROI position receives a canonical binary decision, and*

$$\Omega = \mathcal{A}_\Delta^{\text{can}}(I, \Omega) \dot{\cup} (\Omega \setminus \mathcal{A}_\Delta^{\text{can}}(I, \Omega)).$$

Corollary 8 (Row separability and exterior independence). *If two images have identical active row signal P_y for a row y , their row masks are identical. If $I|_\Omega = J|_\Omega$, then*

$$\mathcal{A}_\Delta^{\text{can}}(I, \Omega) = \mathcal{A}_\Delta^{\text{can}}(J, \Omega).$$

Corollary 9 (Local decision auditability). *Every canonical pixel label has a finite numerical certificate consisting of its frame anchors and the explicit strict inequality responsible for the decision.*

7 Invariance, Parameter Structure, and Stability

Proposition 4 (Additive intensity invariance). *In an ideal real-valued model without clipping or quantization,*

$$\mathcal{S}_\Delta(P + b) = \mathcal{S}_\Delta(P)$$

for every constant b . Hence $\mathcal{A}_\Delta^{\text{can}}(I + b, \Omega) = \mathcal{A}_\Delta^{\text{can}}(I, \Omega)$.

Proof. An additive shift preserves all order comparisons and all intensity differences. It therefore preserves blockers, directional drops, running minima, peaks, frame anchors, and the midpoint and boundary inequalities. $\square$

Proposition 5 (Positive affine photometric equivariance). *For $a > 0$, again in the ideal no-clipping/no-quantization model,*

$$\mathcal{S}_{a\Delta}(aP + b) = \mathcal{S}_\Delta(P).$$

Consequently,

$$\mathcal{A}_{a\Delta}^{\text{can}}(aI + b, \Omega) = \mathcal{A}_\Delta^{\text{can}}(I, \Omega).$$

Proof. Positive affine transformation preserves intensity order and multiplies every intensity difference by a . Simultaneously scaling Δ by a therefore preserves every directional and state comparison. Local normalized coordinates are unchanged because the factor a cancels. $\square$

Proposition 6 (Translation equivariance). *If the image and ROI are translated by the same integer displacement τ within the image domain, then*

$$\mathcal{A}_\Delta^{\text{can}}(\tau I, \tau \Omega) = \tau(\mathcal{A}_\Delta^{\text{can}}(I, \Omega)).$$

Theorem 8 (Peak-regime mask monotonicity). *Let $0 < \Delta_1 < \Delta_2$. If*

$$\mathcal{P}_{\Delta_1}(P) = \mathcal{P}_{\Delta_2}(P),$$

then

$$\mathcal{S}_{\Delta_2}(P) \subseteq \mathcal{S}_{\Delta_1}(P).$$

Regular-frame selected sets are identical at the two thresholds; any shrinkage occurs in boundary frames.

Proof. Equal peak sets imply equal frames. The regular predicate of theorem 5 contains no direct occurrence of Δ , so regular selections are unchanged. In a fixed boundary frame, the threshold $\max\{1/2, \Delta/D_B\}$ is nondecreasing in Δ , so selected boundary pixels can only disappear. $\square$

Corollary 10 (Pixel gain requires a structural transition). *If $\Delta_1 < \Delta_2$ and a pixel is absent from $\mathcal{S}_{\Delta_1}(P)$ but present in $\mathcal{S}_{\Delta_2}(P)$, then*

$$\mathcal{P}_{\Delta_1}(P) \neq \mathcal{P}_{\Delta_2}(P).$$

Proposition 7 (Finite contrast regimes). *For a finite row, every Δ -dependent Boolean comparison changes only at an intensity-difference value. Thus the canonical mask is piecewise constant on the connected components of*

$$(0, \infty) \setminus \{P(i) - P(j) : P(i) > P(j)\}.$$

For 8-bit intensities there are at most 256 unit contrast regimes. If $\Delta \geq 255$, the canonical mask is empty.

Proof. Every contrast comparison has the form $P(i) - P(j) > \Delta$. With finite data there are finitely many distinct left-hand sides. For 8-bit data every positive difference is an integer in $\{1, \dots, 255\}$. If $\Delta \geq 255$, no strict contrast test can succeed; hence no regular peak is accepted, and the whole-row boundary also has dynamic range at most 255 and therefore selects no pixel. $\square$

Proposition 8 (Tie-free horizontal reflection equivariance). *If all intensities of P are distinct and P^{rev} denotes the horizontally reversed row, then*

$$\mathcal{S}_{\Delta}(P^{\text{rev}}) = \text{rev}(\mathcal{S}_{\Delta}(P)).$$

Proof. With no ties, theorem 1 reduces peak selection to the bilateral score π , which is symmetric under exchanging left and right. All interval minima are unique and therefore reflect to the corresponding minima of the reversed interval. The regular and boundary classification predicates depend only on the same reflected anchor intensities, so the final mask reflects exactly. $\square$

Proposition 9 (Quantitative local stability). *Assume a tie-free row and define an ordering margin*

$$\gamma_{\text{ord}} = \min_{i \neq j} |P(i) - P(j)| > 0.$$

Let $\gamma_{\Delta} > 0$ be the minimum distance from zero of all canonical comparison quantities $P(i) - P(j) - \Delta$ actually used by the execution, and let $\gamma_{\text{mid}} > 0$ be the minimum absolute regular/boundary midpoint score $|2P(x) - P(a) - P(b)|$ used in the final frame interpretation. If $\tilde{P} = P + \eta$ and

$$\|\eta\|_{\infty} < \min \left\{ \frac{\gamma_{\text{ord}}}{2}, \frac{\gamma_{\Delta}}{2}, \frac{\gamma_{\text{mid}}}{4} \right\},$$

then

$$\mathcal{S}_{\Delta}(\tilde{P}) = \mathcal{S}_{\Delta}(P).$$

Proof. Any pairwise order or pairwise contrast changes by at most $2\|\eta\|_{\infty}$, so the first two margins preserve all strict ordering, blocker, contrast, and state decisions and hence preserve the frame structure. A midpoint score $2P(x) - P(a) - P(b)$ changes by at most $4\|\eta\|_{\infty}$, so the third margin preserves all midpoint signs. Boundary contrast checks are already covered by γ_{Δ} . $\square$

8 Computational Complexity and an Output-Preserving Linear-Time Reformulation

8.1 Reference directional work

The quadratic worst case does not arise from multiple complete-row or complete-image traversals. The outer procedure traverses the active row positions once. The additional cost arises because one leftward and one rightward directional search are initiated at each position, and the intervals visited by searches originating at different positions may overlap. On a constant row, neither a strictly greater blocker nor a contrast witness exists. The total directional work is therefore

$$\Theta \left(\sum_{p=S}^E ((p-S) + (E-p)) \right) = \Theta(n^2).$$

Frame interpretation itself is linear in the row length.

8.2 Augmented monotone stacks

For the right side, use a right-to-left stack. Each stack entry stores a pair (i, μ_i) , where i is a block anchor and μ_i is the minimum intensity in the contiguous block represented by that entry. When processing p , pop while the top anchor satisfies $P(i) \leq P(p)$. Equal values are popped because the canonical blocker relation is strict: an equal-height sample is not a blocker. After the pops, the surviving top, if any, is the nearest strictly greater blocker $g_R(p)$. The popped blocks concatenate to the interval $(p, g_R(p))$, so the minimum of their stored minima is exactly the bounded minimum needed by the right directional test. A symmetric left-to-right pass computes the left quantities. Figure 4 summarizes the right-pass geometry.

Lemma 4 (Augmented-stack interval invariant). *Immediately before processing p in the right-to-left pass, the stack represents a disjoint contiguous partition of the already processed suffix. Each entry stores the minimum over its represented block and an anchor whose value is the maximum within that block. Popping all anchors with value at most $P(p)$ merges exactly the consecutive blocks in $[p+1, g_R(p))$.*

Proof. The invariant is immediate for the first processed position. Inductively, every popped block has anchor value at most $P(p)$ and hence contains no value greater than $P(p)$. The first surviving anchor is therefore the nearest strictly greater sample. The popped blocks are consecutive by the stack partition invariant, and adding the singleton $\{p\}$ creates the new block $[p, g_R(p))$. Its minimum is the minimum of $P(p)$ and the popped block minima, and its maximum is $P(p)$. $\square$

Lemma 5 (Exact directional recovery). *The two augmented-stack passes recover exactly the same nearest strictly greater blockers and bounded minima as the canonical per-position directional searches, for every row position.*

Proof. The interval invariant identifies the first strictly greater blocker exactly. The union of popped blocks is precisely the open interval between the candidate and that blocker, so taking the minimum of the stored block minima gives the same lowest sample that the directional search could encounter before termination. The left pass is the mirror image. $\square$

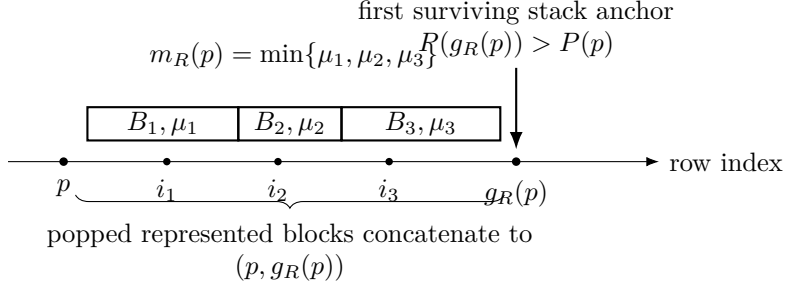


Figure 4: Schematic of right-directional recovery by the augmented monotone stack. Entries with anchor value at most $P(p)$ are popped; their represented blocks concatenate exactly to the interval before the nearest strictly greater blocker. The minimum of their stored block minima is therefore the bounded minimum required by the canonical right directional test.

Theorem 9 (Exact output-preserving linear-time reformulation). *For every finite admissible row P and every $\Delta > 0$, let $\mathcal{S}_\Delta^{\text{fast}}(P)$ be obtained by replacing only the canonical per-position directional searches with the two augmented-stack passes, while retaining the canonical running-minimum state, frame construction, and frame interpretation. Then*

$$\mathcal{S}_\Delta^{\text{fast}}(P) = \mathcal{S}_\Delta(P).$$

Consequently,

$$\mathcal{A}_\Delta^{\text{fast}}(I, \Omega) = \mathcal{A}_\Delta^{\text{can}}(I, \Omega).$$

The reformulated procedure runs in $\Theta(n)$ time per row when an explicit row mask is produced, and in $\Theta(|\Omega|)$ time for an explicit ROI mask. If a full $H \times W$ output image is materialized, the overall bound is $\Theta(HW)$.

Proof. By exact directional recovery, both procedures compute identical left and right Boolean outcomes at every position. Induct on the outer row traversal. Before each position, the two procedures have identical running minima and identical partial frame structures. They update the running minimum under the same strict rule, evaluate the same complete peak predicate, and therefore either both reject or both accept the current position. On acceptance, both close and start the same frames and reset state at the same point. Thus the accepted peaks and frame families are identical. Since frame interpretation is unchanged, every final pixel decision is identical.

For complexity, each position is pushed onto each monotone stack once and popped at most once, so both directional passes are linear. The retained state/frame traversal is linear, as is explicit frame interpretation. An explicit n -pixel row output also gives an $\Omega(n)$ lower bound, hence $\Theta(n)$. $\square$

Corollary 11 (Contrast-independent geometry preprocessing). *For a fixed row, the nearest-greater blockers, bounded minima, and bilateral scores $\pi(p)$ are independent of Δ and can be computed once in $O(n)$ time. In the tie-free case this single preprocessing encodes the entire peak hierarchy through*

$$\mathcal{P}_\Delta(P) = \{p : \pi(p) > \Delta\}.$$

For an ROI with row lengths n_y , the canonical directional stage has worst-case complexity

$$\Theta\left(\sum_y n_y^2\right),$$

whereas the reformulation is $\Theta(\sum_y n_y) = \Theta(|\Omega|)$ for explicit ROI output. For a full-width ROI, these specialize to $\Theta(HW^2)$ and $\Theta(HW)$, respectively.

9 Reference Software Implementation and Step-by-Step Demonstration

The canonical procedure analyzed above originates from a web-based software system developed by the author in 2019. The historical application uses PHP for server-side image processing and JavaScript for browser-side interaction. Rather than presenting the software only as a separate implementation note, this section follows the processing sequence step by step and uses historical figures from that software project to connect ROI specification, row-signal construction, extrema/frame processing, and the displayed pre-revision output. The archived application belongs to the broader historical software lineage and contains later revision layers; the formal operator of this paper remains the pre-revision procedure defined in section 2.

9.1 Step 1: image upload and ROI specification

After image upload, the user draws the region of interest in the browser and invokes processing after closing the ROI boundary. Figure 5 shows the historical interface sequence. In the notation of this paper, the resulting region is denoted by Ω . The implementation-level ROI mechanism and the admissible-ROI assumption of section 2 should be distinguished: the interface permits a user-drawn region, whereas the formal analysis assumes that every analyzed nonempty row slice is represented by one contiguous active interval.

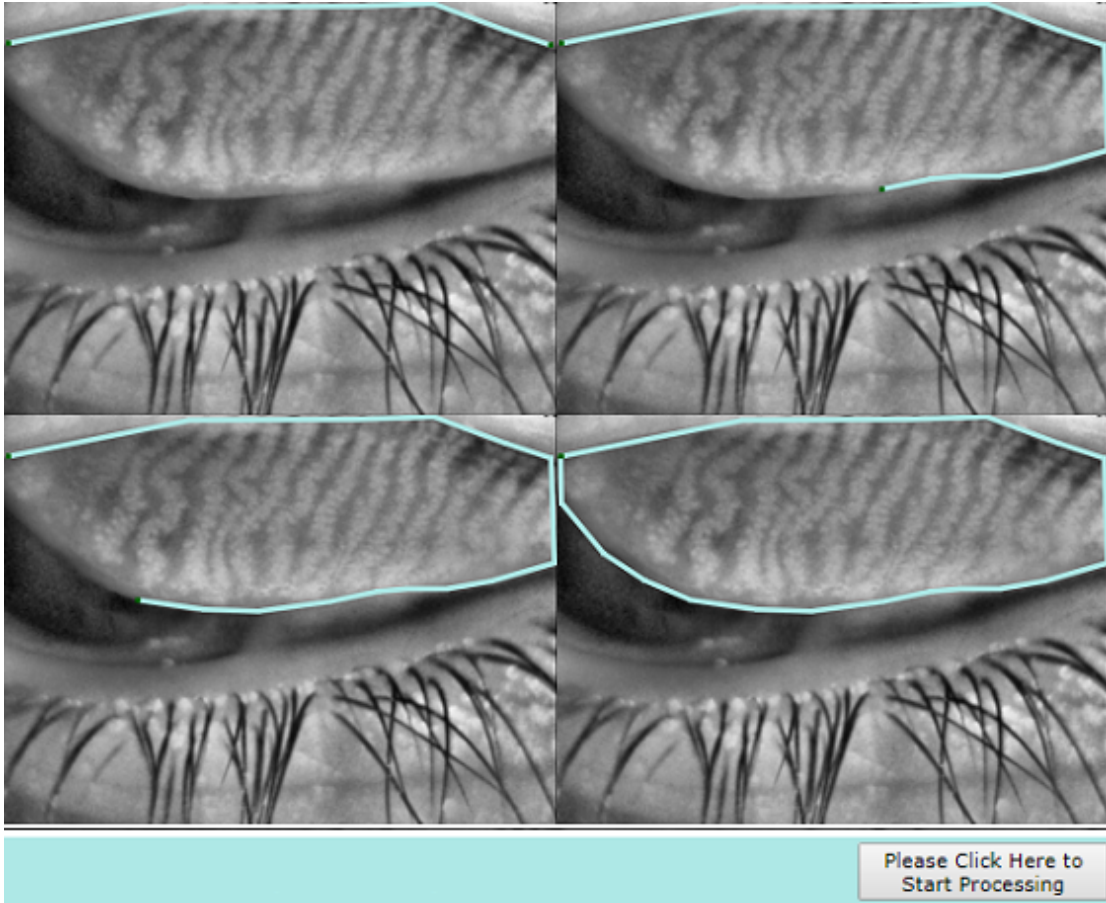


Figure 5: Historical ROI-specification workflow (thesis Figure 4.2). The image is uploaded, the ROI boundary is drawn in the browser, and the resulting active region is sent for processing.

9.2 Step 2: row-wise intensity representation

Inside the active region, the server processes the image row by row. The original description uses a line-chart interpretation of each row: the software need not display a chart during execution, but the grayscale values along the active row form the one-dimensional signal P_y analyzed in section 2. The original thesis presents two distinct image-to-profile examples; both are retained here so that the subsequent extrema and frame illustrations can be read against more than one row morphology.

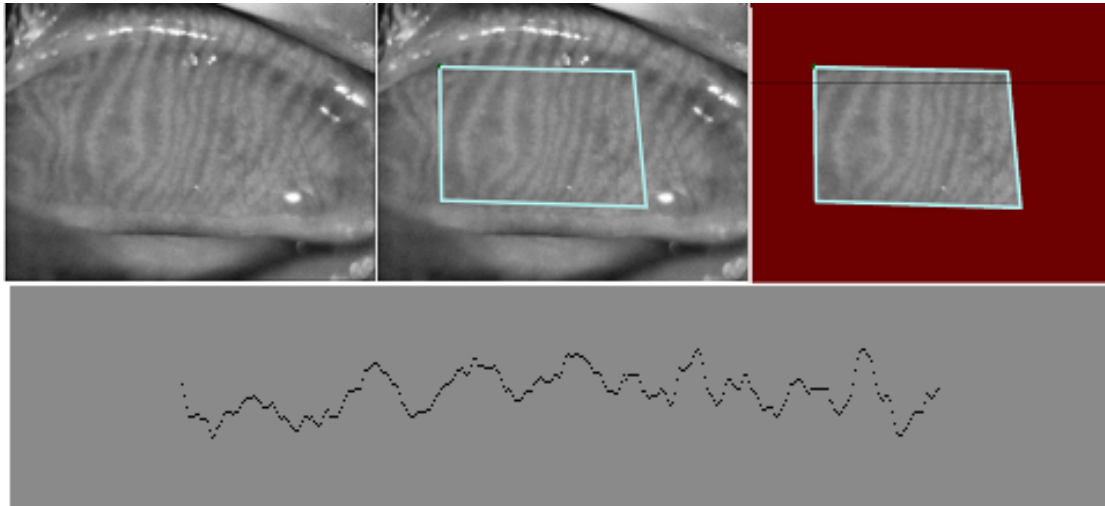


Figure 6: First historical image-to-row-profile example (thesis Figure 3.9). The lower panel visualizes the one-dimensional grayscale signal associated with row 138 of the selected region.

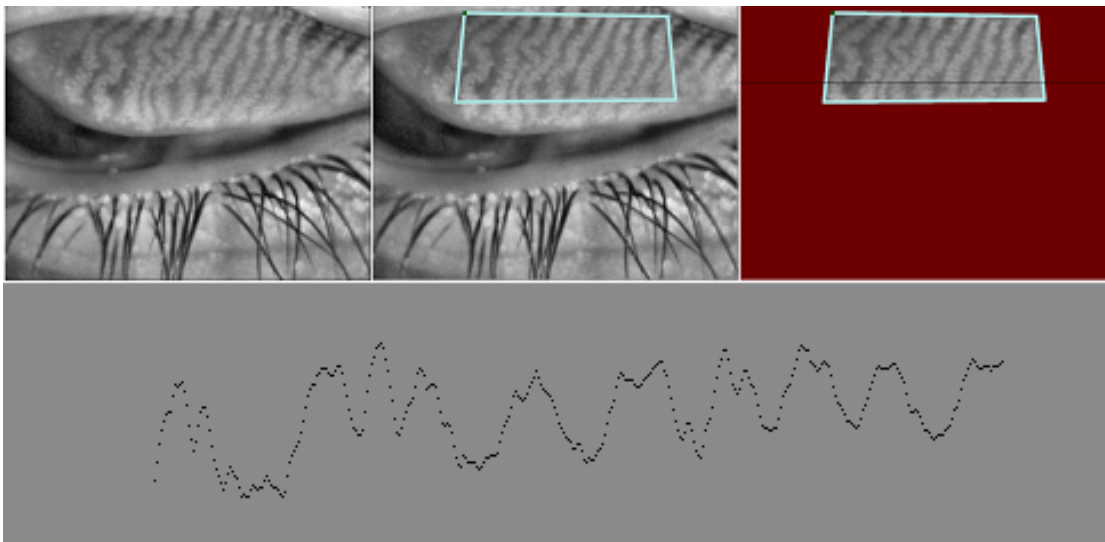


Figure 7: Second historical image-to-row-profile example (thesis Figure 3.10), included to show the same row-signal construction on a different meibography image and signal morphology.

9.3 Step 3: potential regions, extrema, and frame construction

The row profiles are next interpreted in terms of candidate bright regions and extrema. Figure 8 reproduces the thesis illustration in which potential frame regions are marked on a row signal.

This figure is retained because it provides the visual bridge from the bare one-dimensional signal to the frame-based representation analyzed formally in section 4.

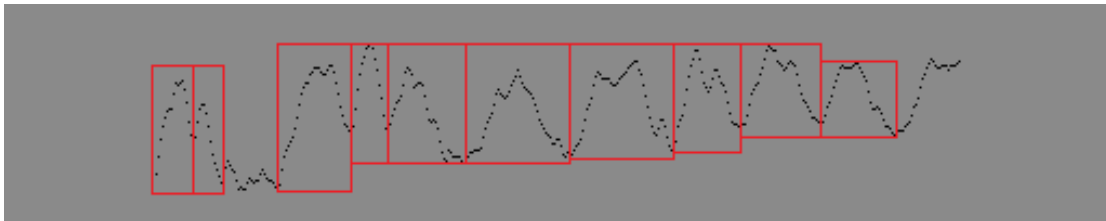


Figure 8: Potential frame regions marked on a historical row signal (thesis Figure 3.12). The marked regions motivate the subsequent minimum–maximum–minimum frame representation.

For each current position of the row, the canonical procedure initiates one leftward and one rightward directional search and maintains the running minimum. This is a single outer traversal with per-position directional searches; it should not be interpreted as multiple complete-row traversals. Positions satisfying the canonical contrast conditions become accepted extrema, and the accepted peaks together with running minima determine the frame sequence. The two historical extrema/frame illustrations in fig. 9 show the maxima/minima markings and the boundary behavior used in the original explanation. The formal prominence-state characterization of this step is developed in section 3, and the unique frame representation is proved in section 4.

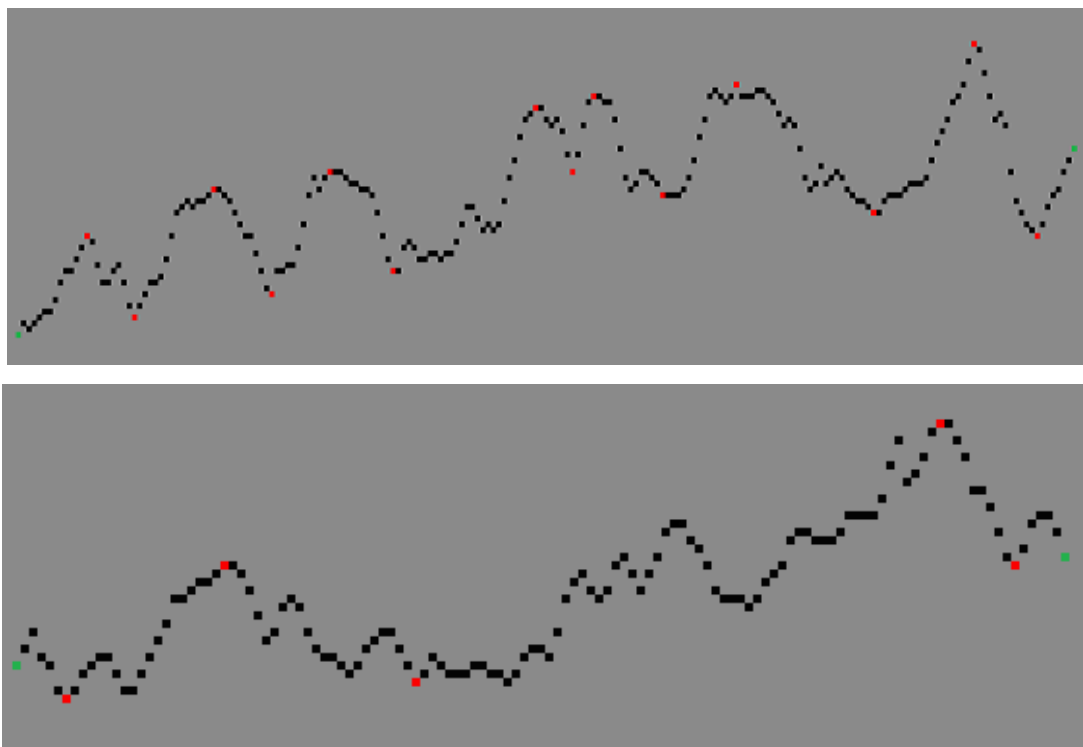


Figure 9: Historical maxima/minima and frame-anchor visualizations. The upper image is the illustration reused in thesis Figures 3.13 and 3.15; the lower image is the illustration reused in thesis Figures 3.14 and 3.16. Red markers indicate extrema, while the endpoint markers support the boundary-frame cases.

9.4 Step 4: frame interpretation and automatic mask generation

After the frames are formed, their pixels are interpreted by the canonical regular- or boundary-frame rule. In the formal notation of this paper, the automatic output is

$$G = \mathcal{A}_{\Delta}^{\text{can}}(I, \Omega).$$

Figure 10 shows the historical pre-revision frame-based pixel-determination result corresponding to thesis Figure 3.17. This figure directly illustrates the output of the frame interpretation stage analyzed in section 5.

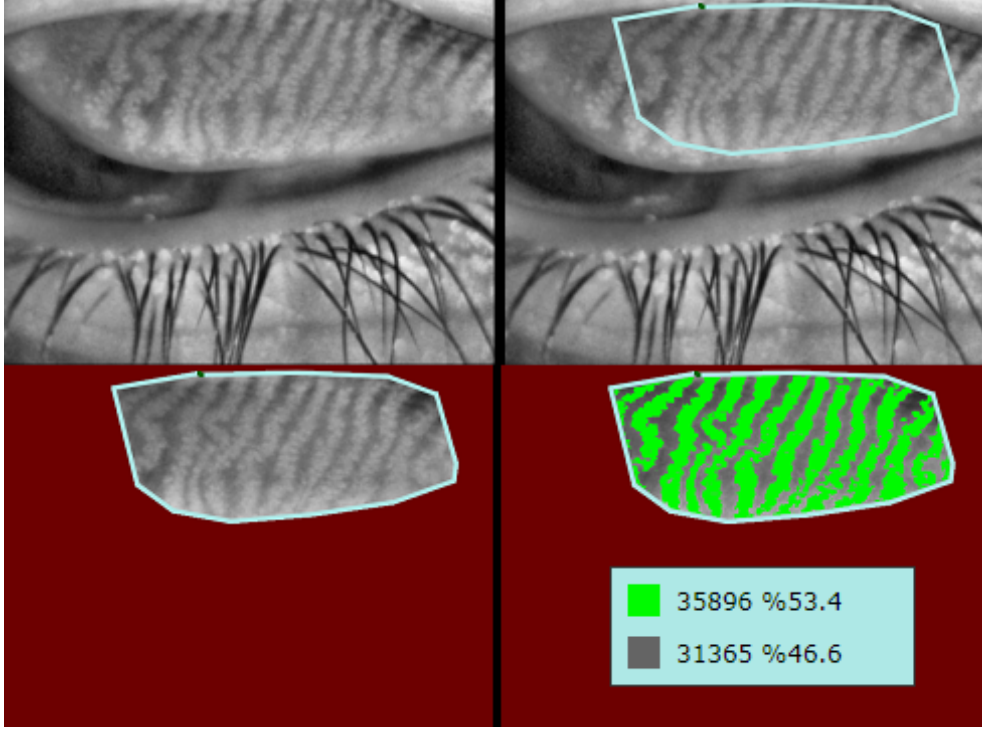


Figure 10: Historical example of automatic frame-based pixel determination within a selected ROI (thesis Figure 3.17). The green overlay represents pixels selected by the pre-revision frame interpretation.

9.5 Step 5: pre-revision software output and mask summaries

For the final historical output shown in this paper, we use thesis Figure 3.18 rather than the later Figure 4.3. Figure 3.18 appears immediately before the thesis section that introduces width-, gap-, and brightness-related algorithm revisions and is therefore the appropriate visual endpoint for the pre-revision lineage studied here. Figure 11 shows the processed ROI together with the displayed class counts and proportions. In the formal notation used here, the direct summaries of the canonical mask are

$$\begin{aligned} N_G &= |G|, & N_{\bar{G}} &= |\Omega| - |G|, \\ r_G &= \frac{|G|}{|\Omega|}, & r_{\bar{G}} &= 1 - r_G. \end{aligned}$$

These quantities summarize the generated mask; they do not introduce another segmentation rule or constitute a clinical-validation result.

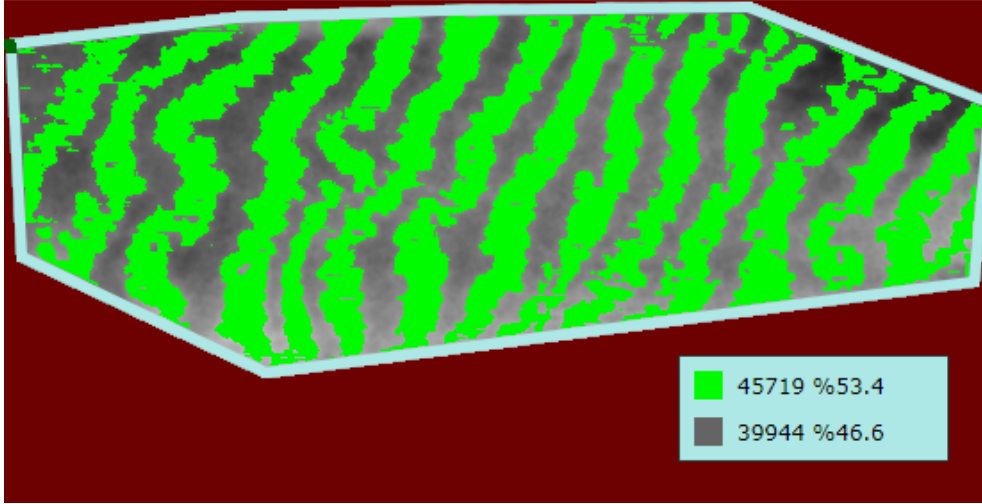


Figure 11: Historical software output immediately preceding the later revision section (thesis Figure 3.18). The processed ROI is displayed together with class counts and proportions. This is used as the visual endpoint of the pre-revision pipeline analyzed in this paper.

9.6 Historical editing and revision layers

The historical interface also permits post-segmentation editing. Such user modifications occur after automatic mask generation and are not part of $\mathcal{A}_{\Delta}^{\text{can}}$. Likewise, the archived software contains later width-, gap-, and brightness-related revision mechanisms. Those mechanisms are deliberately excluded from the operator of section 2, from all theorem statements, and from the output-preservation result. In particular, later software outputs are not used as illustrations of the canonical pre-revision endpoint.

9.7 Historical software versus the present reformulation

The historical software demonstration should not be interpreted as an implementation of the new augmented-stack method. Algorithm 2 changes the computation of the directional geometry while preserving the remaining canonical state, frame-construction, and interpretation semantics. The exact equality proved in theorem 9 is therefore

$$\mathcal{A}_{\Delta}^{\text{fast}} = \mathcal{A}_{\Delta}^{\text{can}},$$

not an assertion that every later revised configuration of the archived web application is pixel-for-pixel identical to Algorithm 2.

9.8 Software demonstration video

A recorded demonstration of the historical web-based workflow is available at

<https://youtu.be/4ach0mB9wu8>.

The recording is presented as a demonstration of the historical software and user workflow, not as clinical validation or as evidence for the mathematical theorems.

10 Discussion and Scope

The analysis shows that the procedural complexity of the canonical segmentation method conceals a compact mathematical structure. Its extrema stage separates into contrast-dependent

bilateral prominence geometry and a deterministic refinement of equal-height recurrence. The accepted extrema induce a unique shared-boundary interval representation, while the original pixel-distance rule reduces to an explicit locally normalized classifier. Thus the method is not only deterministic at the implementation level; its intermediate representation and individual output decisions admit direct mathematical characterization.

The parameter Δ plays two distinct roles. It controls the survival of extrema and therefore the structural resolution of the frame representation, while in boundary frames it also acts directly as a pixel-level contrast safeguard. These roles explain why accepted peaks form a monotone hierarchy although the final segmentation mask need not be globally monotone. Within a fixed peak regime, regular-frame decisions remain unchanged and boundary decisions can only disappear as Δ increases. Pixel gains at a larger Δ therefore imply a change in the underlying peak/frame structure.

The computational reformulation preserves this interpretation. It does not replace the canonical segmentation logic with an approximation or a new classifier. Instead, it computes the directional information required by that logic through augmented monotone stacks and then retains the canonical state, frame-construction, and pixel-interpretation semantics. The resulting reduction from quadratic worst-case directional work to linear row processing therefore comes with exact equality of the canonical output.

A further property is auditability. For fixed image data, ROI, and contrast parameter, every intermediate decision and final pixel label is uniquely determined. A regular-frame decision can be traced to its minimum and peak anchors and the midpoint inequality; a boundary decision can additionally be traced to its explicit Δ -contrast requirement. These are specification-level guarantees and should not be confused with external clinical accuracy.

The scope of the guarantees is intentionally narrow. Correctness means correctness with respect to the canonical operator defined in section 2; output preservation means exact equivalence to that operator. The perturbation result is a conditional computational stability statement away from ties and decision boundaries. The theory excludes later revision layers and post-segmentation editing and assumes contiguous active row slices.

11 Conclusion

This work provides a formal and computational analysis of a deterministic extrema-frame segmentation procedure originally developed for meibography images. We showed that its bilateral extrema search admits a prominence-based characterization and that the apparent state dependence of peak selection reduces to a deterministic treatment of equal-height recurrence. Accepted peaks uniquely determine a four-type shared-boundary frame representation whose evolution with the contrast parameter forms a merge-only laminar hierarchy.

The frame-interpretation stage has an exact normalized form. Regular half-frames use the universal decision $\alpha > 1/2$, while boundary frames obey $\alpha > \max\{1/2, \Delta/D\}$. These characterizations yield explicit guarantees concerning uniqueness, specification correctness, deterministic output, parameter structure, local auditability, and conditional perturbation stability.

We further derived an output-preserving linear-time reformulation of the canonical procedure. Two augmented monotone-stack passes recover the directional geometry obtained by the canonical per-position bidirectional searches. Retaining the remaining state and frame semantics gives exactly the same extrema, frames, and final mask while reducing the worst-case processing time from quadratic to linear per row.

Together, these results turn the original pre-revision procedure into a precisely specified algorithmic object with explicit structural, interpretive, and computational guarantees.

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Conflict of Interest

The author declares no conflict of interest.

A Negative Results and Counterexamples

A.1 The final mask is not globally monotone in Δ

Consider

$$P = (1, 0, 3, 2).$$

Under the canonical rules,

$$\mathcal{S}_{0.5}(P) = \{0, 2\}, \quad \mathcal{S}_1(P) = \{2, 3\}.$$

Neither set contains the other. Thus the final mask is not globally monotone in Δ . This does not contradict theorem 8, because the accepted-peak/frame structure changes between the two thresholds.

A.2 General horizontal reflection equivariance fails with ties

Consider

$$P = (0, 2, 0, 3, 0), \quad \Delta = 2.$$

The canonical mask is $\{1, 3\}$. For the reversed signal

$$P^{\text{rev}} = (0, 3, 0, 2, 0),$$

the canonical mask is $\{1\}$, whereas the geometric reflection of $\{1, 3\}$ is again $\{1, 3\}$. Hence unrestricted horizontal-reflection equivariance is false. The tie-free result in section 7 is therefore genuinely conditional.

B Implementation-Level Equivalence Audit

The formal equivalence proof in theorem 9 is primary. As a separate implementation-level sanity check, two independent evaluators were implemented: a direct canonical evaluator using the per-position directional procedure and an augmented-stack evaluator using the reformulation of section 8. Both implement the same strict pre-revision frame-classification rules.

All signals

$$P \in \{0, 1, 2, 3\}^n, \quad 1 \leq n \leq 8,$$

were enumerated, giving

$$4^1 + \dots + 4^8 = 87,380$$

row signals. For the seven positive thresholds

$$\Delta \in \{0.25, 0.5, 1, 1.5, 2, 2.5, 3\},$$

this yields 611,660 complete row/threshold executions. The two implementations agreed exactly on accepted peaks, regular frames, boundary frames, and final masks in all 611,660 executions.

The augmented-stack blocker/minimum geometry also agreed with direct bounded searches on all 87,380 row signals.

The same exhaustive set produced 1,834,980 ordered threshold-pair checks for accepted-peak monotonicity and the no-boundary-drift property, with no counterexample. Peak-boundedness and regular-frame contrast were checked on 486,756 regular frames. Among threshold pairs with identical accepted-peak sets, 839,016 mask-inclusion checks supported theorem 8. Finally, all $1! + 2! + \dots + 8! = 46,233$ tie-free permutations of lengths one through eight were tested at the seven thresholds, producing 323,631 horizontal-reflection checks with no counterexample.

These finite tests are not proofs of the theorems and are not clinical validation. Their purpose is to audit consistency between the formal specification and independent implementations over an exhaustive finite test space.